

Nifty100 Analytics Platform

Analyst Guide

Version: Sprint 6 Release | Dataset: 92 Nifty 100 Companies

1. Platform Overview

The Nifty100 Analytics Platform is a Python-based financial analytics platform covering 92 companies from the Nifty 100 dataset. It combines financial statement analysis, ratios, cash-flow intelligence, machine-learning clustering, NLP insights, automated reporting, dashboards and REST APIs.

2. Core Analytical Modules

Module	Purpose
Financial Ratios	Profitability, leverage, efficiency and coverage metrics
CAGR	Historical growth analysis
Peer Analysis	Comparison of companies within sectors
Valuation	P/E, P/B, EV/EBITDA and dividend metrics
Capital Allocation	CapEx, working capital, debt and cash deployment
Cash-Flow Intelligence	CFO quality, FCF, conversion and CapEx intensity
Clustering	KMeans-based financial archetype classification
NLP	Rule-based pros, cons and confidence scoring
Screener	Investment-oriented filtering presets
Reporting	Company tear sheets and sector reports
REST API	Programmatic access to analytical results
Dashboard	Interactive Streamlit/Plotly analytics

3. Cash-Flow Intelligence

The cash-flow module evaluates the quality and sustainability of cash generation using CFO/PAT, free cash flow, FCF conversion, CapEx intensity, cash-flow risk and related metrics.

Metric	Interpretation
CFO / PAT	Cash generation relative to reported profit
FCF	Cash remaining after capital expenditure
FCF Conversion	Conversion of accounting profit into free cash flow
CapEx Intensity	Capital expenditure relative to operating scale
CFO Quality	Classification of operating cash-flow strength
Cash-Flow Risk	Flag identifying potential cash-generation concerns

4. Machine-Learning Clustering

KMeans clustering is used with five clusters and a fixed random state of 42. The clustering pipeline assigns all 92 companies to financial archetypes after preprocessing and missing-value handling.

Cluster	Archetype
0	High-Leverage / Cash-Flow Stressed
1	Low-Leverage / Cash-Generative Compounders
2	Capital-Efficient High-ROCE Companies
3	Leverage-Heavy / Weak Cash Conversion
4	Capital-Intensive Profit Reinvestment

5. NLP Intelligence

The NLP pipeline parses financial analysis records and generates company-level pros and cons using predefined financial rules. Statements are assigned confidence scores and integrated into company tear sheets.

6. Automated Reports

Report	Expected Output
Company Tear Sheets	92 HTML reports
Sector Reports	10 HTML reports
Radar Charts	Company-level analytical visualizations
Portfolio Reports	Portfolio-level analytical summaries
Pytest Report	Automated test execution report

7. REST API

The FastAPI service exposes analytics through versioned API routes covering companies, screening, sectors, peers, valuation, portfolio statistics, documents and health.

Area	Purpose
Companies	Company-level analytical information
Screener	Preset and filter-based company screening
Sectors	Sector-level analytics
Peers	Peer comparison
Valuation	Valuation metrics
Portfolio	Portfolio-level statistics
Documents	Report/document access
Health	Service and database health

8. Testing

The project uses pytest for automated validation across ETL, KPI, analytics and API functionality. The current verification run completed with 67 passed tests and one dependency deprecation warning.

9. Key Project Commands

Command	Purpose
make load	Load/refresh data
make ratios	Generate analytical ratios
make test	Run automated tests
make report	Generate reports
make dashboard	Launch dashboard workflow
make api	Launch API workflow
make clean	Clean generated artifacts

10. Output Structure

Important generated outputs include company_clusters_v4.csv, cashflow_intelligence_v4.csv, pros_cons_generated.csv, company tear sheets, sector reports, radar charts, portfolio reports and pytest_report.html.

11. Analyst Interpretation Rules

- Monetary values are interpreted in INR Crore where applicable.
- Financials should be treated carefully when applying generic debt/equity screening logic.
- Negative-base CAGR situations should be interpreted as turnaround cases rather than ordinary CAGR.
- Zero interest expense should be interpreted as debt-free for relevant reporting logic.
- Simulated datasets must be explicitly labelled SIMULATED.
- NLP statements are ranked by confidence score.
- Analytical outputs should be validated before investment interpretation.

12. Current Release Status

The current project contains 92 generated company tear sheets, 10 sector reports and an automated test suite with 67 passing tests. Final Sprint 6 sign-off should additionally verify the required documentation and acceptance-gate evidence.